# Social Media Content Analyzer

Side-by-Side Comparison: Your Strategy Doc vs. Recommended Approach

Context: 3-day window (submitted July 11, due July 14) for the Unthinkable Solutions take-home assignment. Comparison reflects what's explicitly required in the assignment email versus assumptions layered on in the strategy doc.

## 1. Scope, Deployment & Structure

| Area | Your Strategy Doc | Recommended Approach | Verdict |
| --- | --- | --- | --- |
| Time framing | Strategy doc treats it as an over-engineered production build; original framing ignored the 8-hour line in the email entirely. | 3-day window justifies a fuller build, but still triaged: build the core pipeline deeply, defer nice-to-haves to a documented 'future work' list. | Partial match |
| Deployment | Two services: React+TS+Tailwind on Vercel, FastAPI backend on Render. | Same split is fine now that there's more time to test it, but a single-service deploy (FastAPI serving a lightweight frontend) is still lower-risk if time gets tight. | Your call, tested |
| Project structure | Full separation: frontend/ (components, pages, hooks, services, utils) and backend/ (api, services, models, utils, config). | Same shape is reasonable with 3 days. Recommend keeping backend services modular but resisting extra abstraction layers that add files without adding clarity. | Keep, trim gently |
| Core framing assumption | "Not primarily an AI assignment, it's an SWE assessment disguised as an AI project." | Treat this as an unverified hypothesis, not a certainty. Hedge by making the AI/analysis output genuinely solid, since under-investing there is the one mistake that's hard to recover from. | Risky as stated |

## 2. Analysis Output & AI Component

| Area | Your Strategy Doc | Recommended Approach | Verdict |
| --- | --- | --- | --- |
| Output fields | 8 fields: Engagement Score, Sentiment, Tone, Readability, Strengths, Weaknesses, Missing CTA, Hashtag suggestions, Improved version. | Pick 4-5 that are most defensible and make them genuinely good: Sentiment, Engagement Score, Strengths/Weaknesses, Improved version. Add 1-2 more only if the core four are solid with time to spare. | Trim scope |
| Structured output | Correctly specifies returning structured output instead of raw LLM text. | Agreed as-is. Use a strict Pydantic schema server-side so the LLM response is always coerced into a validated shape before returning to the client. | Keep as-is |
| LLM choice | Gemini 2.5 Flash (free tier). | Reasonable choice given free-tier constraints and email's 'any free tier AI service' permission. No change needed. | Keep as-is |

## 3. Edge Cases & Testing

| Area | Your Strategy Doc | Recommended Approach | Verdict |
|---|---|---|---|
| Edge case count | 14 edge cases listed for full handling (unsupported types, corrupted PDFs, size limits, blank files, OCR fallback, mixed PDFs, no text found, LLM timeout, network failure, duplicates, malformed OCR, long text, unicode/emoji, multi-post parsing). | Implement the 5-6 an evaluator will actually test directly: wrong file type, image-only PDF triggering OCR, no text extracted, oversized upload, API/LLM failure. Document the rest as acknowledged limitations in the README rather than fully coding all 14. | Trim to core 5-6 |
| Test suite | Full 25-test suite specified. | 10-12 targeted tests mapped 1:1 to the documented edge cases reads as more deliberate than 25, which can look padded. | Trim scope |
| Documentation checklist | 10-item checklist: architecture, setup, env vars, design decisions, assumptions, limitations, future improvements, screenshots, deployment link, API description. | All reasonable with 3 days, but keep total README length tight and lead with a ~200-word approach summary that mirrors the write-up you'll submit separately. | Keep, keep tight |

## Bottom Line

Your strategy doc's instincts (structure, docs, edge-case awareness) are sound. The main correction is triage: with 3 days, build the core upload → extract → OCR fallback → LLM analysis → structured output pipeline until it's genuinely reliable, narrow the analysis fields and edge-case handling to what's defensible, and turn everything you're not implementing into clearly labeled README limitations rather than silently dropping it. A tight, working product with an honest limitations section outperforms a sprawling one with gaps an evaluator finds by accident.